Outbreak Zero

Build a fictional outbreak simulation using a 2D list. Calculate simultaneous turns correctly, then investigate immunity and hospital placement using repeated trials.

Complete the prep HTML first. Allow 5–7 lessons of 90 minutes, or longer if needed. Finish all six stages; save a working version after each.

Stage 1 Store and display a small world

Do this

1. Use the supplied 5×5 test grid. Store states 0 empty, 1 susceptible, 2 infected and 3 immune in world[row][column]. Indices start at 0.

2. Create independent rows. Changing one cell must not change another row.

3. Display . H X I with a key. Write reusable display and state-count functions.

4. Return only up/down/left/right neighbours within the grid. No diagonals or wrapping.

5. Check corners, edges and centre cells before adding spread. Population excludes empty cells.

Check before moving on

The supplied small grid has 5 empty, 16 susceptible, 1 infected and 3 immune cells: population 20. A corner has 2 neighbours, an edge 3 and a centre 4.

Watch out

Rows that share the same list object can change together. A diagonal cell is not a neighbour in this model.

Stage 2 Generate a reproducible 20 by 20 world

Do this

1. Generate independent 20×20 rows. Each cell has a 70% occupancy chance; occupied cells start susceptible.

2. Choose one occupied cell as infected. If the grid is empty, create a person first.

3. Assign immunity to the chosen percentage of remaining occupied people. Exclude the initial infected person and state your rounding rule.

4. Accept immunity, transmission and recovery percentages from 0–100 and a whole-number seed. Reject invalid input and retry. Defaults: transmission 25%, recovery 10%.

5. Record requested immunity, actual immune count and starting population. Fresh runs with the same seed and settings must repeat.

Check before moving on

Two fresh runs with the same seed and settings begin identically. At 0% immunity none of the remaining people are immune; at 100% all remaining people are immune but the initial case remains infected.

Watch out

A 70% occupancy chance does not guarantee exactly 280 people. Immunity is assigned among occupied people other than the initial case.

Stage 3 Calculate one simultaneous turn

Do this

1. Predict [2,1,1] with 100% transmission and no recovery: one turn gives [2,2,1].

2. Read decisions from the old grid; write changes into a new grid with copied, independent rows.

3. Each old infected person tries each old susceptible neighbour independently. Convert percentages to probabilities.

4. Count each newly infected target once, even if several attempts succeed.

5. After transmission, apply recovery to old infected people only; recovered people become immune. New cases wait until next turn to transmit or recover.

6. Return the new grid and new-case count. Keep empty and immune cells unchanged; do not alter the input grid.

Check before moving on

The row test gives [2,2,1]. With both transmission and recovery at 100%, it gives [3,2,1]. At 0% transmission there are no new cases. Your function does not alter its input grid.

Watch out

Writing into the grid you are reading can let infection cross several cells in one turn. Copying only the outer list still shares its rows.

Stage 4 Run an outbreak and measure its outcome

Do this

1. Repeat simultaneous turns. Stop at zero infected or 100 completed turns; report the stopping reason.

2. Track current infection, peak simultaneous infection and cumulative people ever infected. Include the initial case in peak and cumulative.

3. Add each turn's unique new cases to cumulative. Recovered people remain part of this total.

4. An epidemic means cumulative infection is at least 25% of the initially occupied population.

5. Add a repeating menu for fresh runs. Save seed, settings, population, duration, peak, cumulative and epidemic outcome. Displaying every grid is optional.

Check before moving on

A persistent initial case at 0% transmission and 0% recovery stops at turn 100 with peak 1 and cumulative 1. At 100% recovery and 0% transmission, one initial case ends after one turn.

Watch out

Peak and cumulative are different measurements. Reaching the turn limit does not mean the infection has disappeared.

Stage 5 Investigate immunity using repeated trials

Do this

1. Save 100 different seeds. Run 100 fresh trials for each immunity level: 0%, 20%, 40%, 60%, 80%. Reuse the same seed list across levels.

2. Hold grid size, occupancy chance, transmission, recovery and 100-turn limit fixed.

3. Save every trial's seed, settings, population, initial immune count, duration, peak, cumulative and epidemic outcome.

4. Calculate mean peak, mean cumulative and epidemic frequency: epidemic trials ÷ 100 × 100%. Also compare cumulative as a percentage of starting population.

5. Export raw trials and the five-row comparison. Explain your numerical findings and why random outcomes vary.

Check before moving on

You have 500 trial records, five summary rows and the seed list. Repeating the experiment with the same implementation and settings reproduces its results. Every mean uses 100 trials.

Watch out

One attractive animation is not enough evidence. Resetting to one identical seed 100 times gives repeated copies of one trial.

Stage 6 Compare hospital placement and submit evidence

Do this

1. Store hospitals as separate coordinates, without replacing population cells.

2. Halve transmission into a susceptible target beside any hospital, using orthogonal adjacency. Apply the reduction once; recovery is unchanged.

3. At one fixed immunity level, compare no hospitals, two central hospitals and two edge hospitals over the same 100 seeds.

4. Use central (9,9)/(10,10) and edge (0,9)/(19,10), or justify alternatives. Record coordinates and keep other settings fixed.

5. Export 300 hospital trials and their comparison. Explain any small differences cautiously.

6. Submit code, README, tests, seed list, all 800 trial records, both comparisons, findings and a 200–300 word reflection: actual bug, fix, two tests, limitation and help used.

Check before moving on

The full investigation contains 500 immunity trials and 300 hospital trials. Your tests include boundaries, no cascade, unique new cases, 0/100% probabilities, seed repeatability and the turn limit.

Watch out

Protection depends on the susceptible target, not the infected source. Two adjacent hospitals do not halve the chance twice.